

ELECENG 2E14: Electronic Devices and Circuits 1

Design Project 1

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Summary

The objective of this project is to design and implement a low-voltage dc power supply capable of delivering 10 mA at $3.0\text{ V} \pm 0.1\text{ V}$ from a 120 V (rms), 1 kHz ac source. The power supply follows the standard AC-to-DC conversion process, consisting of a transformer for voltage reduction, rectification, filtering, and optional regulation. The AC input voltage is first reduced using a transformer, then converted into a unipolar signal using a rectifier. A capacitor-based filter is used to reduce voltage ripple by storing energy during voltage peaks and releasing it during voltage drops, resulting in a smoother dc output. A voltage regulator may be used to further improve output stability by maintaining a constant voltage under varying load conditions.

For safety reasons, the 120 V source is not used directly in the design implementation. Instead, a function generator is used to generate the analytically determined secondary voltage of the transformer. In this design, regulation is achieved through careful selection of rectifier and filter components rather than an active regulator. Based on design calculations, simulation, and experimental measurements, the circuit delivers an output voltage close to 3.0 V at the required 10 mA load current, with voltage ripple maintained within the specified tolerance.

Design

Transformer

A transformer is not physically used in the design implementation; however, the circuit is designed assuming that the input voltage is derived from the secondary voltage of a transformer connected to a 120 V (rms), 1 kHz ac source. The role of the transformer in this design is to step down the high input voltage to a safe level suitable for the rectifier and filter stages.

To determine the required transformer secondary voltage, the voltage losses in the rectifier and filter stages must be considered. A full-wave bridge rectifier is used in the design, which results in two diode forward voltage drops during conduction. Assuming a forward voltage drop of 0.7 V per diode, the total rectifier voltage drop is approximately 1.4 V. In order to obtain a dc output voltage of approximately 3.0 V at the load after filtering, the peak voltage at the rectifier input must therefore exceed 4.4 V.

Relationship between peak and rms voltage for sinusoidal input:

$$V_{peak} = \sqrt{2} V_{rms}$$

Based on this formula, if we choose $V_{rms} = 3.2\text{V}$, we will get $V_{peak} = 4.525\text{ V}$, which provides sufficient margin to account for diode voltage drops and voltage ripple while maintaining the required output voltage.

If this 3.2 V (rms) input were generated from a 120 V (rms) ac source, the required transformer turns ratio would be:

$$E_p = \text{primary voltage}$$

$$E_s = \text{secondary voltage}$$

$$\frac{E_p}{E_s} = \frac{120}{3.2} = 37.5 : 1$$

Thus, a transformer with a 37.5:1 primary-to-secondary turns ratio would be required to produce the desired secondary voltage.

Rectifier

A full-wave bridge rectifier topology is used for this design to convert ac input voltage into unipolar output. This configuration uses 4 diodes arranged such that both positive and negative half of input sinusoidal waveform to contribute to the output which helps reduce ripple, produce smoother DC with higher efficiency and smaller filter capacitor as compared to half cycle rectifier which uses only one half of AC cycle resulting in pulsating DC with gaps and low efficiency.

During each conduction cycle, two diodes are forward-biased, resulting in a total forward voltage drop of approximately 1.4 V, assuming a forward voltage drop of 0.7 V per diode. This voltage drop was accounted for when selecting the transformer secondary (rectifier input) voltage. The full-wave bridge rectifier was chosen because it offers better performance with minimal additional complexity and does not require a center-tapped transformer, making it well-suited for low-voltage power supply applications.

The diode used in the rectifier has the following parameters:

- Forward Voltage Drop, $V_F = 0.7 \text{ V}$
- The maximum reverse breakdown voltage V_{BR} is to be significantly greater than the peak input voltage
- Maximum forward current rating: above the load current of 10 mA

1N4148 diodes are used in this design

Filter

To reduce the voltage ripple present at the output of the rectifier, a capacitor-input filter is used. The filter consists of a single capacitor connected in parallel with the load at the output of the rectifier. This filter charges the capacitor when the diodes are conducting and discharges that energy when the rectifier diodes are not conducting, which helps gives more accurate DC current. The required capacitor is determined based on the allowable ripple voltage and load current.

For a full-wave rectifier, the ripple frequency is twice the input frequency. Given an input frequency of 1 kHz, the ripple frequency is 2 kHz.

It is given that the DC output voltage should be $3.0 \text{ V} \pm 0.1 \text{ V}$, which means

$$V_{rpp(max)} \leq 0.2 \text{ V}$$

Using the RC filter formula:

$$I_{out} = I_{load} = 10 \text{ mA} = 0.01 \text{ A}$$

$$f = f_{ripple} = 2000 \text{ Hz}$$

$$V_{rpp} = 0.2 \text{ V}$$

$$V_{rpp} = \frac{I_{out}}{fC}$$

$$C = \frac{I_{load}}{f_{ripple} V_{rpp}} = \frac{0.01 A}{2000 Hz \times 0.2 V} = \frac{0.01}{400} = 2.5 \times 10^{-5} F = 25 \mu F$$

The minimum capacitance needed is $25 \mu F$.

Ripple voltage is inversely proportional to capacitance, so if $C > 25 \mu F$ then,

$$V_{rpp} = \frac{I_{out}}{fC} < 0.2 V$$

We don't have a $25 \mu F$ capacitor in the kit, so we can use $100 \mu F$ as it will reduce ripple even more.

$$V_{rpp} = \frac{I_{out}}{fC} = \frac{0.01}{2000 \times 100 \times 10^{-6}} = \frac{0.01}{0.2} = 0.05 V \text{ (this fulfils the requirement of } C \geq 25 \mu F \text{)}$$

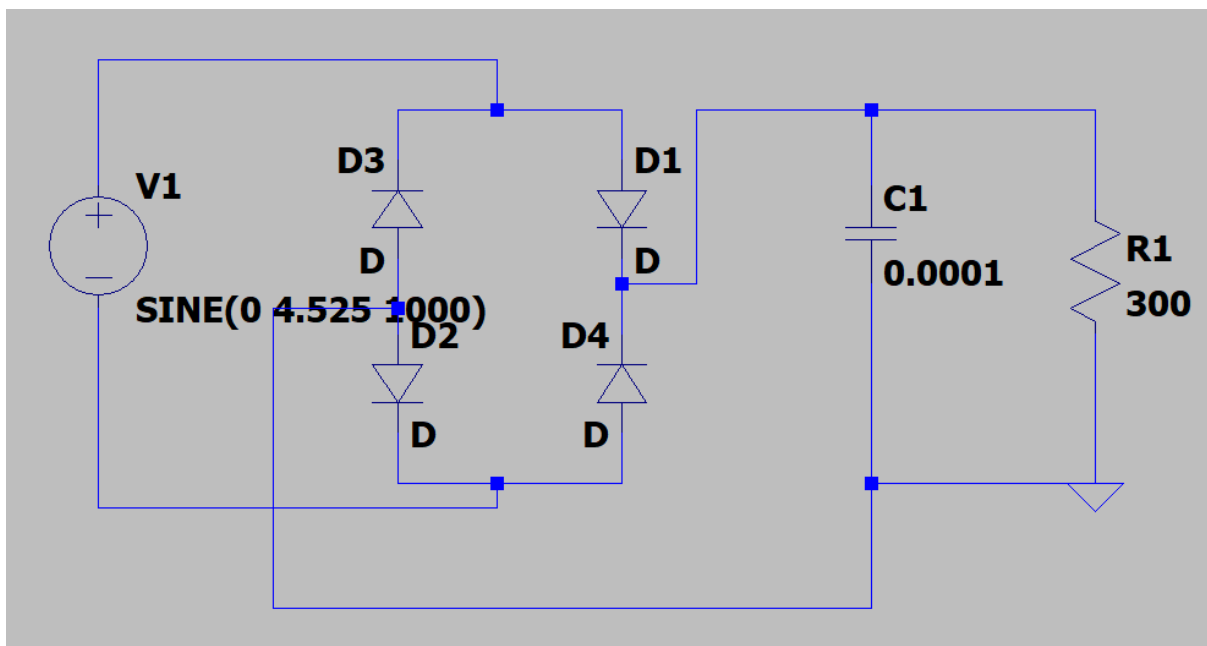
and $V_{rpp} \leq 0.2 V$

This means the output will vary between $\pm 0.025 V$ of the output voltage

Regulator

A regulator has not been used in this design.

Schematic



Summary of calculations and expected performance

Load Resistor

DC output required (given): $V_{DC} = 3.0 V$

Load Current (given): $I_{load} = 10 mA = 0.01 A$

$$\text{Using Ohm's Law: } R_L = \frac{V_{DC}}{I_{load}} = \frac{3.0}{0.01} = 300\Omega$$

Calculations for turn ratio (30:1), transformer secondary value(4V rms), minimum capacitance required(25 μ F), selected capacitance(100 μ F) and expected ripple voltage for this design(0.05V) have been done in previous sections.

Expected Performance (detailed calculations have been done in the section above)

- | | |
|---|--|
| a. Input Condition | $V_{peak} = 4.525 \text{ V}$ |
| b. After the bridge Rectifier | $V_{peak, rect} = 4.525 - 1.4 = 3.125 \text{ V}$ |
| c. Ripple Voltage using 100 μ F a capacitor | $V_{rpp} = 0.05 \text{ V}$ |
| d. Average DC Output | $V_{DC} = V_{peak, rect} \pm \frac{V_{rpp}}{2} = 3.125 \pm 0.025 \text{ V}$ |
| e. Expected Output Range | $V_{max} = 3.125 \text{ V}$
$V_{min} = V_{max} - V_{rpp} = 3.125 - 0.05 = 3.075 \text{ V}$
$3.075 \text{ V} < V_{out} < 3.125 \text{ V}$ |

Design Tradeoffs

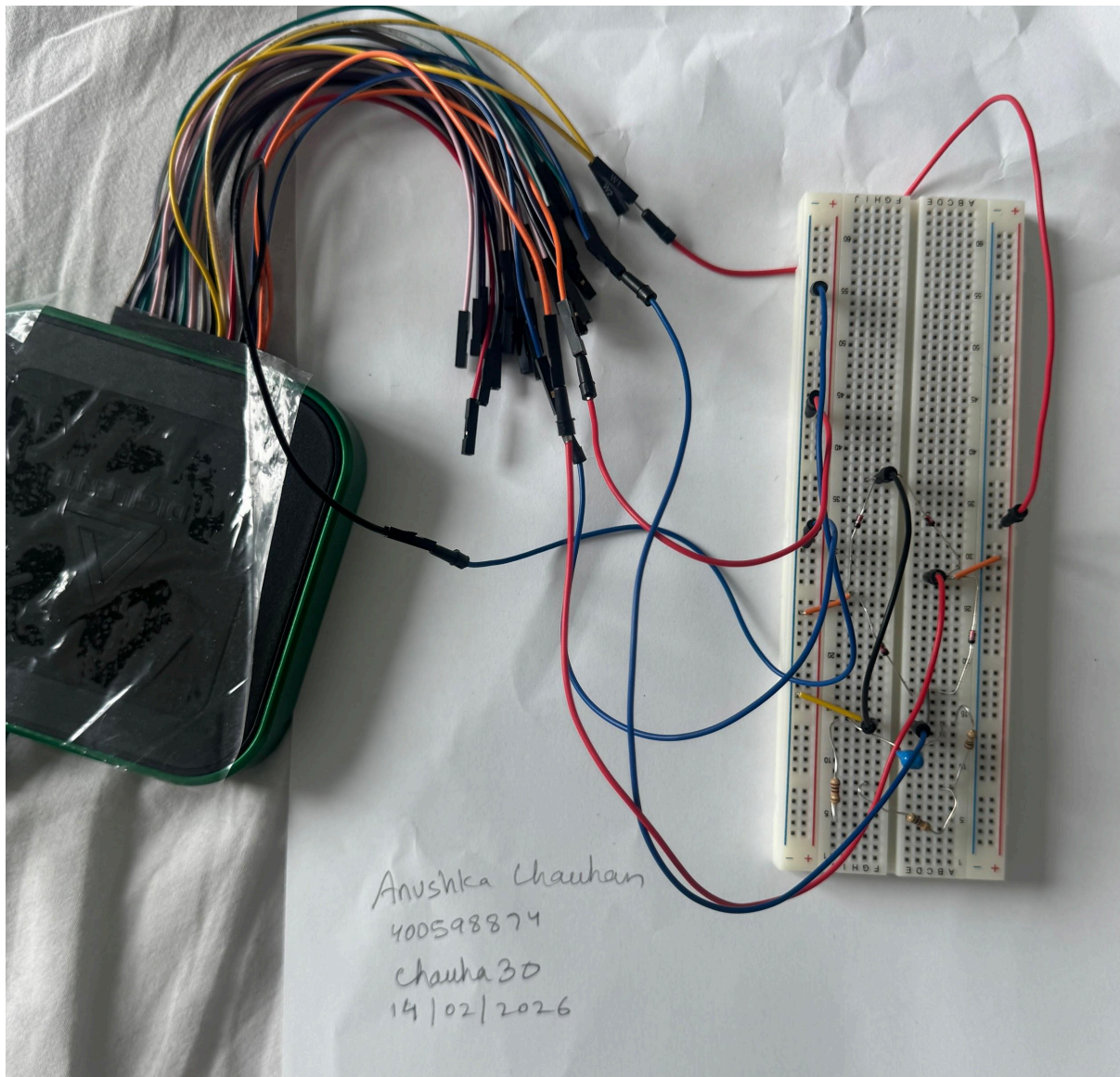
In designing this AC-to-DC power supply, several tradeoffs and design decisions were carefully considered. A full-wave bridge rectifier was chosen instead of a half-wave rectifier because it uses both halves of the AC cycle, which reduces ripple and allows the use of a smaller capacitor. The tradeoff is that two diodes conduct at the same time, causing a slightly larger voltage drop, but this was acceptable since reducing ripple was more important for meeting the output specification. The capacitor size was also a tradeoff: larger capacitance reduces ripple but increases physical size, cost, and inrush current at startup. A 100 μ F capacitor was selected, which is higher than the minimum required, providing a safety margin and ensuring the ripple stays well below the specification of $\pm 0.1 \text{ V}$. The decision not to use a voltage regulator simplified the circuit and reduced cost, but it also means the output voltage depends directly on the input and load, making it less stable under varying conditions.

The 1N4148 diodes were chosen because their reverse voltage and forward current ratings far exceed the expected peak voltage ($\sim 4.5 \text{ V}$) and load current ($\sim 10 \text{ mA}$), providing a large safety margin. The capacitor voltage rating (35 V) is much higher than the maximum output, ensuring reliable operation. The load resistor, rated at 0.25 W, safely handles the power dissipation of only 0.03 W, again providing a strong safety margin. The circuit uses a low-voltage AC source from the AD3 rather than directly from 120 V mains, eliminating the risk of electric shock, overheating, or damaging components. All components are operated well below their maximum ratings to ensure safe lab operation.

Overall, the design balances simplicity, safety, and performance. By carefully selecting the rectifier, capacitor, and other components with appropriate ratings and margins, the circuit achieves the required output voltage and ripple while ensuring safe and reliable operation in the lab environment.

Measurement and Analysis

Circuit



Measurement Procedure

The performance of the rectifier circuit was evaluated using the Digilent Analog Discovery 3 (AD3). A sinusoidal input voltage was generated using the Wavegen function with a peak amplitude of 4.53 V, zero DC offset, and a frequency of 1 kHz. This corresponds to approximately 3.2 V RMS, which matches the transformer secondary voltage selected in the design calculations. The generated AC signal was applied to the input terminals of the bridge rectifier. The rectified output was connected to a 100 μ F filter capacitor and a 300 Ω load resistor, implemented using three 100 Ω resistors connected in series.

To evaluate performance, Oscilloscope Channel 1 was connected across the AC input terminals to verify the applied sinusoidal waveform and confirm that the correct amplitude

and frequency were being supplied. Oscilloscope Channel 2 was connected across the load resistor (which is in parallel with the filter capacitor) to measure the DC output voltage and observe the ripple voltage. The maximum and minimum output voltages were determined directly from the oscilloscope waveform to calculate the ripple and average DC level.

In addition, a Math custom function was used on the oscilloscope to estimate the output current through the load resistor. Since the load resistance is $300\ \Omega$, the current was calculated using Ohm's law as $I_{out} = \frac{V_{out}}{300\Omega}$. In the oscilloscope math channel, this was implemented using the expression C2/300, where C2 represents the measured output voltage waveform. This allowed direct visualization of the load current waveform and verification that the expected output current (approximately 10 mA for a 3 V output) was achieved.

Measurement Result

Measured Voltage Output Values from AD3:

- Maximum Output Voltage (V_{max}) = 3.10 V
- Minimum Output Voltage (V_{min}) = 2.90 V
- Ripple Voltage(V_{epp}) = $V_{max} - V_{min} = 0.20\text{ V}$

This satisfies the design requirement of $2.9\text{ V} < V_{out} < 3.1\text{ V}$

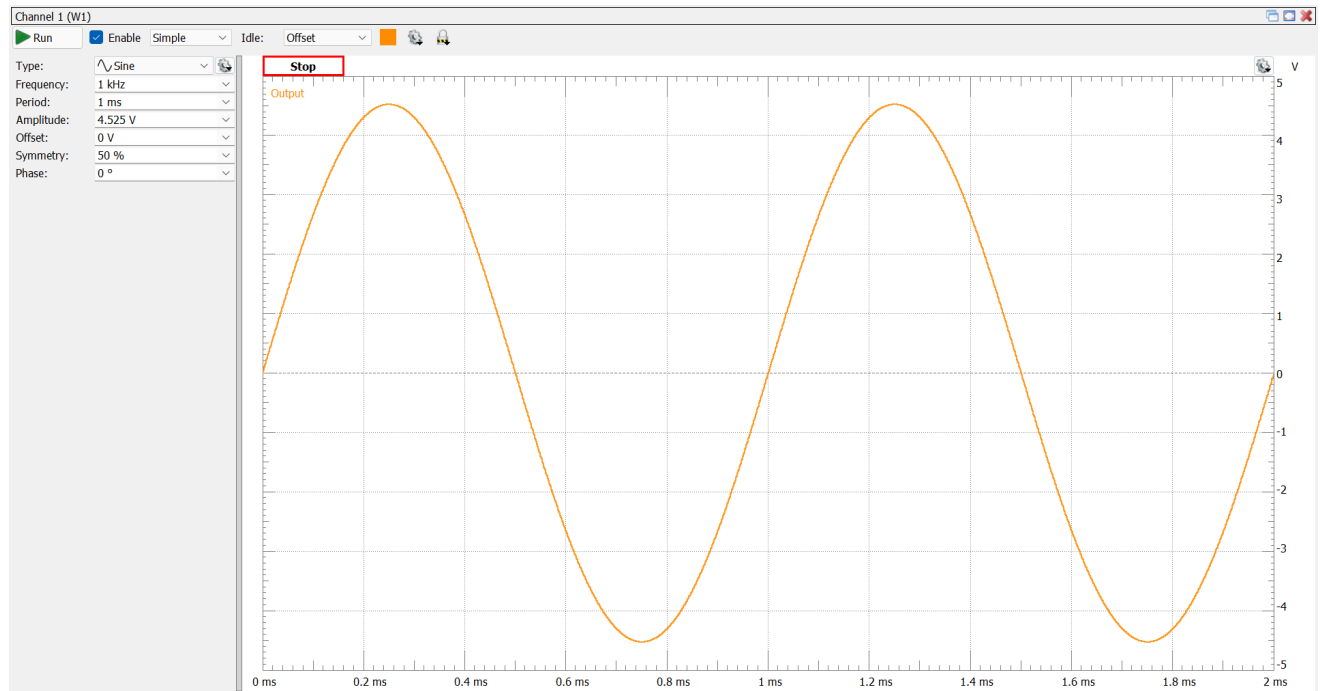
Measured Current Output Values from AD3 using math function:

- Maximum Load Current = 10.5 mA
- Minimum Load Current = 9.5 mA

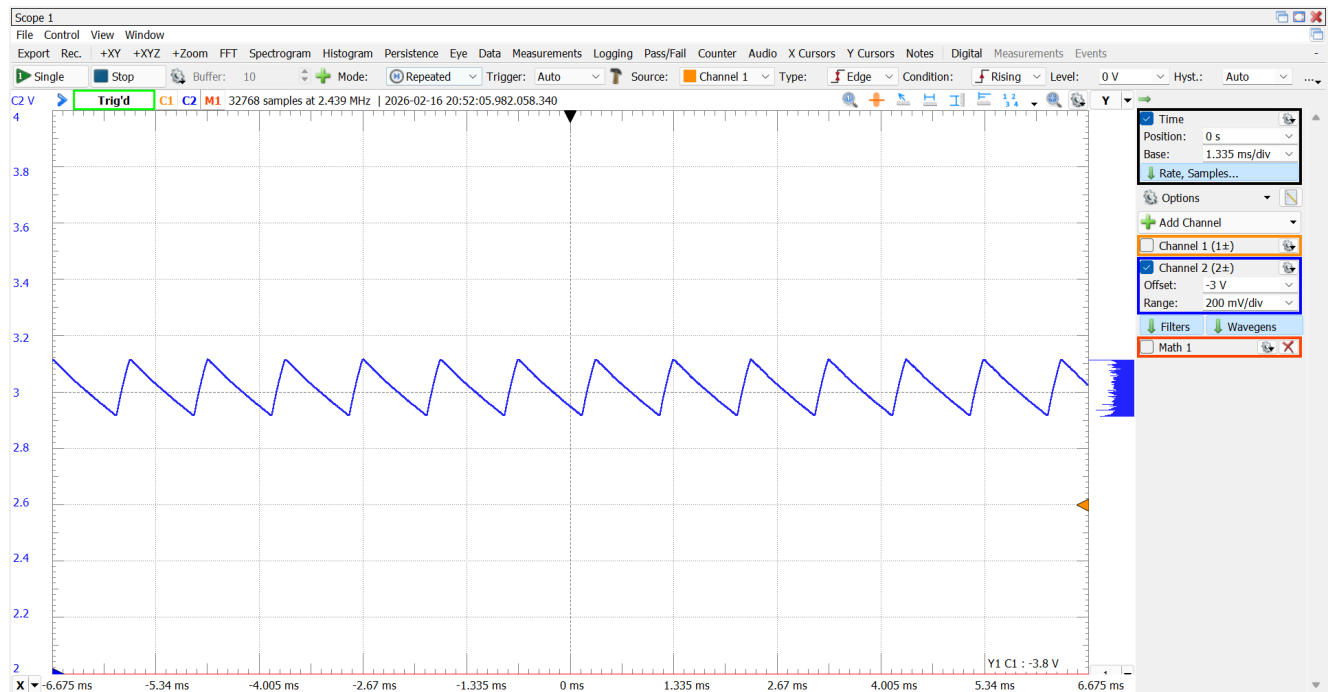
The measured current values closely match theoretical expectations, confirming that the rectifier and filter circuit are operating according to design specifications.

Oscilloscope Output

- a. Voltage Waveform supplied as input to the circuit



- b. Voltage Waveform received as output at the load

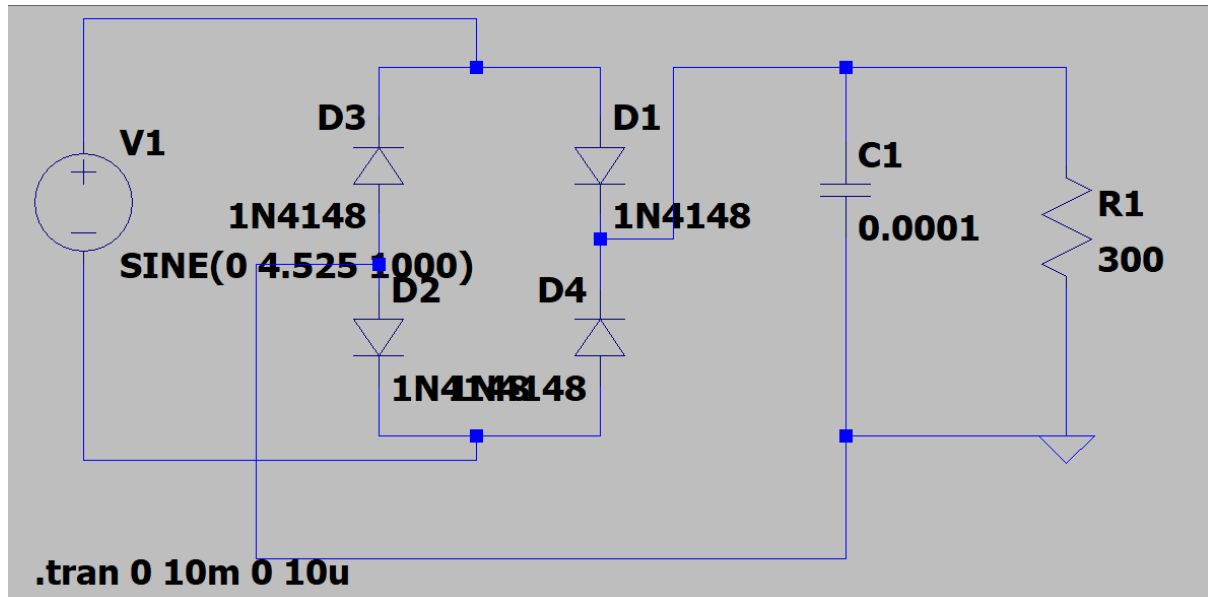


c. Current waveform received as output at the load



Simulation

Circuit Schematic



Netlist

```
* C:\Users\camit\Documents\LTspice\Draft2.asc
* Generated by LTspice 26.0.1 for Windows.
V1 N001 N003 SINE(0 4.525 1000)
C1 N002 0 0.0001
R1 N002 0 300
```

```

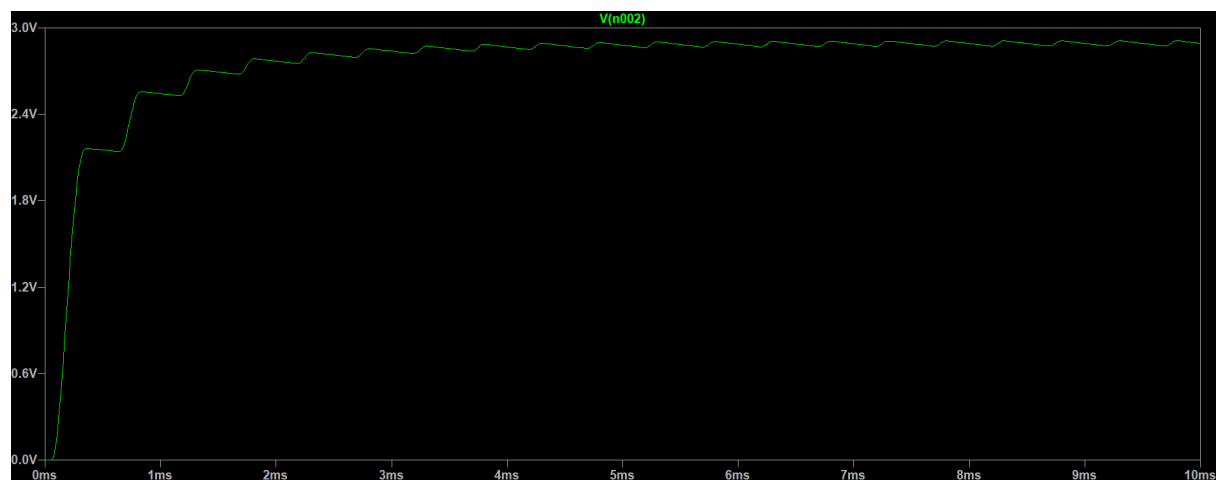
D1 N001 N002 1N4148
D2 0 N003 1N4148
D3 0 N001 1N4148
D4 N003 N002 1N4148
.model D D
.lib C:\Users\camit\AppData\Local\LTspice\lib\cmp\standard.dio
.tran 0 10m 0 10u
.backanno
.end

```

Simulation Conditions

A transient simulation was performed in LTspice to observe how the circuit behaves over time. The input voltage was set as a sine wave using SINE(0 4.525 1000), which means the signal has a peak value of 4.525 V, zero DC offset, and a frequency of 1 kHz. This represents the transformer secondary voltage used in the design. The transient command used was .tran 0 10m 0 10u, meaning the circuit was simulated from 0 to 10 milliseconds. Since the input frequency is 1 kHz, 10 ms allows us to see 10 full cycles of the waveform and ensure the output reaches steady state. The small time step of 10 microseconds was chosen to show the ripple in the output voltage clearly. Transient analysis was used because we want to see how the output voltage changes over time, including the charging of the capacitor and the ripple voltage. The bridge rectifier diodes used in the simulation were modeled as 1N4148 diodes from the LTspice standard diode library.

Simulation Output



The output shows capacitor charging and reaching close to 3.0V, and reaches a steady state in about 3 ms. The steady state output voltage seems to go from 2.8V to 2.9V, which is close to the expected output range of 2.9V to 3.1V.

Discussion

From the hand calculations, the expected DC output voltage of the circuit was approximately 3.125 V, with a small ripple of about 0.025 V. These calculations assumed ideal conditions, such as a constant 0.7 V voltage drop across the diode and an exact capacitor value of 100 μF . In the LTspice simulation, the output voltage reached a steady state after roughly 3 ms and fluctuated slightly between 2.8 V and 2.9 V, reflecting the more realistic behaviour of the diode model used. When measured on the physical circuit using the AD3, the output voltage ranged from 3.10 V (maximum) to 2.90 V (minimum), resulting in a peak-to-peak ripple of 0.20 V. The load current measured between 9.5 mA and 10.5 mA, which is very close to the expected 10 mA. Overall, the measured results meet the design requirement of $2.9\text{ V} < V_{\text{out}} < 3.1\text{ V}$, confirming that the circuit works as intended.

There are small differences between the calculated, simulated, and measured results due to practical factors. Hand calculations assume ideal components, while the simulation uses a diode model that allows the forward voltage drop to vary slightly with current. In the physical circuit, components are not perfect; the capacitor may differ from its nominal 100 μF value and has internal resistance, and the input voltage from the AD3 may vary slightly. Additional factors, such as small wiring resistances and oscilloscope measurement noise, also affect the results.

The design itself has some limitations. Because there is no voltage regulator, the output voltage depends directly on the input voltage and the load current. If the load resistance decreases, the ripple increases and the average output voltage drops, setting a maximum allowable load current to keep the voltage within 2.9 V to 3.1 V. In a rectifier circuit with a capacitor filter, the capacitor charges up to the peak voltage of the rectified AC signal. When there is a load connected, the capacitor discharges through the load between peaks, causing the output voltage to drop slightly before the next peak recharges it. This drop is the ripple voltage. However, with no load, almost no current flows, so the capacitor barely discharges between peaks. As a result, the output voltage stays much closer to the peak voltage of the rectified waveform, and the ripple becomes very small. The capacitor essentially acts like a “storage tank” of voltage: with no current drawn, it stays nearly full, keeping the voltage near its maximum; with a load, the voltage drops slightly as current is drawn before the next peak. Measurement tools also have their own accuracy limits and may introduce small noise.

In conclusion, the circuit operates as expected. The measured voltages and currents are close to the calculated values, and any small differences can be explained by real component behaviour and practical measurement limitations.

References

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